

A 12 year experience of colistin resistance in *Klebsiella pneumoniae* causing neonatal sepsis: two-component systems, efflux pumps, lipopolysaccharide modification and comparative phylogenomics

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Background: Increased use of colistin in healthcare necessitates studies on the trend of colistin resistance and the underlying mechanisms.

Objectives: To understand the susceptibility trend and molecular mechanisms of colistin resistance in neonatal isolates over a 12 year period.

Methods: Colistin susceptibility, mRNA expression, whole genome sequence and mutational analysis was performed. Phylogenomic comparison with a global collection of colistin-resistant *Klebsiella pneumoniae* strains ($n = 70$) was done.

Results: Of 319 Enterobacteriales (*K. pneumoniae* and *Escherichia coli*) studied, colistin resistance was found in 9 *K. pneumoniae* (2.8%). The transmissible colistin resistance gene, *mcr*, was absent. Colistin-resistant *K. pneumoniae* belonged to diverse sequence types (ST14/37/101/14/7/16) and exhibited multiple mechanisms of colistin resistance including overexpression of the two-component systems (TCS) (*phoP/Q*, *pmrA/B*), and AcrAB-TolC pump and its regulators. Mutations in TCS, *mgrB*, pumps, repressors, and lipopolysaccharide-modifying genes were detected. Phylogenomic comparison revealed that this global collection of colistin-resistant *K. pneumoniae* was diverse, with the presence of epidemic and international clones. Mutations in *mgrB* and TCS noted in global strains were comparable to the study strains. Co-occurrence of carbapenem resistance ($n = 61$, 87%) was observed in global strains. Co-existence of dual carbapenemases (*bla*_{NDM-5} with *bla*_{OXA-48/181}) in multiple lineages within different replicons was found in neonatal colistin-resistant study isolates only.

Conclusions: Colistin resistance both in study and global strains is multifaceted and attributed to mutations in chromosomal genes leading to lipopolysaccharide modification or efflux of colistin through pumps. With no transmissible *mcr*, prevalence of colistin-resistant strains was low in this unit. Colistin-resistant strains with dual carbapenemases causing sepsis are alarming as they are practically untreatable.

Introduction

The limited options for treating antibiotic-resistant organisms have led to renewed usage of colistin as salvage therapy. Colistin, a cationic antibiotic, exerts bactericidal activity by destabilizing LPS.¹ Modification of LPS by two-component system (TCS) genes (*phoP/phoQ* and *pmrA/pmrB*), *mgrB* (negative regulator of *phoP/phoQ*), efflux pumps, their regulators, lipid A biosynthesis and assembly genes, all contribute to colistin resistance.¹ A plasmid-mediated colistin resistance gene, *mcr*, has spread successfully across the globe, further worsening the situation.²

Colistin usage is still limited in neonates, but recent studies showed colistin administration in neonates from Southeast Asia

and the Middle East.³ Considering colistin as a ‘last resort’ antibiotic, this study aimed to decipher the trend of colistin susceptibility among septicaemic neonatal strains over a 12 year period, and analyse the different mechanisms of colistin resistance among strains with/or without carbapenem resistance.

Materials and methods

Ethics

Ethics approval (Memo No. IPGME&R/IEC/2020/026) was obtained from the Institutional Ethics Committee prior to the study, maintaining patient confidentiality.

Isolation of colistin-resistant strains and their genomic characterization

Septicaemic neonatal Enterobacteriales (2008–19) ($n=319$) were assessed for colistin susceptibility by broth microdilution (BMD).⁴ Molecular typing was done by PFGE and MLST. WGS was performed using Illumina (paired-end) and submitted to the NCBI database. Further characterization of resistance genes and plasmids was carried out by different online pipelines (see the [Supplementary methods](#) available at JAC Online).

Evaluating mechanisms of colistin resistance

The presence of the *mcr* gene was screened for, by PCR.⁵ To assess efflux pump activity, colistin MIC was determined with the efflux pump inhibitors carbonyl cyanide 3-chlorophenylhydrazone (CCCP; 10 mg/L), 1-(1-naphthylmethyl)-piperazine (NMP; 25 mg/L) and phenylalanine-arginine β -naphthylamide (PABN; 40 mg/L) (Sigma-Aldrich, St Louis, MO, USA) (see [Supplementary methods](#)). mRNA levels of TCS genes (*phoP/phoQ*, *pmrA/pmrB*), efflux pumps (AcrAB-TolC, KpnEF, KpnGH), with their global regulators (*soxS* and *ramA*) were analysed.⁶ Mutations of genes responsible for colistin resistance such as *phoP/phoQ*, *pmrA/pmrB*, *crrA/crrB*, *mgrB*; LPS formation, modification, assembly and export genes along with efflux pumps, their regulators and repressors were studied ([Supplementary methods](#)). Domain architecture and effect of mutations on protein activity were evaluated by Simple Modular Architecture Research Tool (SMART) (<http://smart.embl.de/>) and Protein Variation Effector Analyzer (PROVEAN) (http://provean.jcvi.org/seq_submit.php).

Comparative analysis of global colistin-resistant genomes with focus on genes associated with colistin resistance

Comparative analysis of global colistin-resistant strains (63 isolates from 12 studies with WGS reported, and 7 from this study) was carried out. Genomes were annotated, characterized, and a phylogenetic tree was generated using Roary (v3.12.0) and IQ-TREE (v2.0) ([Supplementary methods](#)). Data analysis was restricted to the mutations observed by the authors of these genomes as analysis of each mutation was out of the scope of this study. Mutations were annotated onto the tree and visualized using iTOL (v5.7).

Results and discussion

Characterization of colistin-resistant strains

Out of 79 *E. coli* and 240 *K. pneumoniae*, only 9 *K. pneumoniae* showed elevated colistin MIC (8–64 mg/L), whereas others were susceptible (0.25–1 mg/L) (Figure [S1A and B](#)). Colistin resistance in the unit was low ($n=9$, 2.8%) and emerged from 2015 onwards. Colistin-resistant *K. pneumoniae* (ColR-Kp) strains Kp1 to Kp9 were isolated from neonates who never received colistin. Eight ColR-Kp (Kp2 to Kp9) are characterized here, as colistin resistance mechanism of the first ColR-Kp (Kp1) has already been published.⁶

Strains were resistant to aminoglycosides, cephalosporins, fluoroquinolones, carbapenems and sulfamethoxazole/trimethoprim and were diverse as per PFGE and MLST. Some of them belonged to international clones (ST14, ST101 and ST147) (Table [1](#) and Figure [S2](#)). Global ColR-Kps are also diverse, implying the absence of any dominant colistin-resistant clone.^{6–13}

ColR-Kp genomes harboured dual carbapenemase (*bla*_{NDM-5} and *bla*_{OXA-48/181}) (Figure [S3](#)) along with different β -lactam

resistance genes on different replicon types (Table [1](#)). *bla*_{NDM-1,5} was associated with ISAb125 (Figure [S4](#)), while *bla*_{OXA-48} and *bla*_{OXA-181} with truncated Tn1999 and Tn2013 respectively (Figure [S5](#)). Occurrence of concomitant carbapenem resistance has been noted in non-neonatal strains from the USA, Mexico, Netherlands, Hungary, Greece, Italy, Spain, France and also in India.^{8,14} This study is probably the first to report presence of dual carbapenemases in colistin-resistant neonatal isolates.

Analysing mechanisms of colistin resistance

mcr genes were absent among the ColR-Kps, and upregulated TCS genes, *phoP* (4–6-fold), *phoQ* (5–9-fold), *pmrA* (4–13-fold) and *pmrB* (6–11-fold) in most strains indicated chromosomal resistance mechanisms (Table [1](#)). All mutations in *phoP* (Ile26Gln, Arg114Ala, Arg128Ser), one in *pmrB* (Thr246Ala) and *pmrC* (Ile138Val) were common among the study strains and non-deleterious as per PROVEAN analysis (Table [2](#)). One mutation noted in *phoP* (Arg128Ser) was not reported earlier while others were similar to mutations described previously.^{1,8,15} Similarly, presence of a stop codon in the regulator *mgrB*, as noted here, was reported previously,¹⁶ while other substitutions in *mgrB* were novel (Cys28Gly, Val32Gly). These mutations were deleterious and present in the main functional domain, which might modulate the protein activity and contribute to colistin resistance. Mutations and constitutive activation of *pmrCAB* and *phoPQ* collectively probably led to activation of the *arnBCADTEF* operon resulting in colistin resistance.^{1,2} PmrAB activator, CrrAB was mutated in the study strains (Table [2](#)) and mutations in *crrAB* confer resistance to colistin in *K. pneumoniae*.²

Deleterious mutations in the *arnBCADTEF* operon (*arnB*, *arnD*, *arnT*) might have induced colistin resistance as previously reported.^{8,13} According to PROVEAN, all the strains have non-deleterious mutations in *ugd* and lipid A biosynthesis genes (*lpxA* and *lpxD*), except *lpxB* (Ala28Glu, Asp251Gly), *lpxM* (Ala66Ser) and *pagP* (Phe187Ile) which had a deleterious mutation (Table [2](#)). The LPS export gene *lptD* (Arg586His) showed deleterious mutation in Kp7. Few ColR-Kps harboured mutations in the *rcs* operon, which modulates capsule biosynthesis (Table [2](#)). Faulty lipid A and defective LPS assembly may have contributed to colistin resistance as demonstrated previously.^{1,2,17}

Few studies have shown role of efflux pumps in colistin resistance^{1,6} and a decreased colistin MIC with CCCP treatment among the study strains indicated involvement of efflux pump, which was corroborated by elevated expression of AcrAB-TolC pump genes (*acrA*, *acrB*, *tolC*), and its regulator, *ramA*, in all ColR-Kps (Table [1](#)). Substitutions were detected in *acrA*, *acrR*, *acrB*, *tolC*, *ramR*, *marR*, *soxS*, *kpnE*, *kpnF*, *macA* and *macB*, of which, *ramR* (Lys9Ile, Ala20Asp) and *macB* (Ala611Val) had deleterious mutations (Table [2](#)). No mutations were detected in KpnGH, *ramA* and *marA*, but a mutated *ramR* and *marR* might influence *ramA* and *marA*, which can modify LPS, contributing further to colistin resistance.^{2,18}

Phylogenomic analysis of ColR-Kp genomes focusing on genes contributing to colistin resistance

Whole genome sequences of ColR-Kps ($n=70$) collected from 2010 onwards from 11 countries in Europe, South America, North America and Asia, including seven genomes from this

Table 1. Genomic characterization and mRNA expression analysis of genes responsible for colistin resistance in *Klebsiella pneumoniae*

Antimicrobial susceptibility					RT-PCR (fold change) ^c							Accession no.
Isolate	ST/CC	Resistance phenotypes ^a	CST MIC (mg/L)	Antimicrobial resistance genes	Plasmid types	TCS	Efflux pumps and their regulators					
							RND pump	Regulators	SMR pump	MFS pump		
EN5339/Kp2	ST14/CC15	AMK, GEN, CTX, FOX, CIP, MEM, SXT	64 R	<i>rmtB</i> , <i>oqxAB</i> , <i>aac(6')-Ib-cr</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{NDM-5}</i> , <i>bla_{OXA-181}</i> , <i>bla_{TEM-1A}</i> , <i>bla_{SHV-28}</i> , <i>bla_{OXA-19}</i>	ColKP3, IncFIA (H11), IncFIB(K), IncFIB (pKPHS1), IncFII, IncFII (K), IncR	<i>phoP</i> (4.6), <i>phoQ</i> (5.4)	<i>acrA</i> (3.18), <i>acrB</i> (3.4), <i>tolC</i> (4)	<i>ramA</i> (5.6)	<i>kpnE</i> (3.5), <i>kpnF</i> (2)	<i>kpnG</i> (3.3), <i>kpnH</i> (2)	VSIJ000000000	
EN5340 ^b /Kp3	ST14/CC15	AMK, GEN, CTX, FOX, CIP, MEM, SXT	64 R	<i>rmtB</i> , <i>oqxAB</i> , <i>aac(6')-Ib-cr</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{NDM-5}</i> , <i>bla_{OXA-181}</i> , <i>bla_{TEM-1A}</i> , <i>bla_{SHV-28}</i> , <i>bla_{OXA-19}</i>	ColKP3, IncFIA, IncFIB, IncFII, IncFII (K), IncR	<i>phoP</i> (4.5), <i>phoQ</i> (5.7)	<i>acrA</i> (3), <i>acrB</i> (3.4), <i>tolC</i> (4)	<i>ramA</i> (5)	<i>kpnE</i> (3.3), <i>kpnF</i> (2.1)	<i>kpnG</i> (3.4), <i>kpnH</i> (2)	WGS ND	
EN5343 ^b /Kp4	ST14/CC15	AMK, GEN, CTX, FOX, CIP, MEM, SXT	64 R	<i>rmtB</i> , <i>oqxAB</i> , <i>aac(6')-Ib-cr</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{NDM-5}</i> , <i>bla_{OXA-181}</i> , <i>bla_{TEM-1A}</i> , <i>bla_{SHV-28}</i> , <i>bla_{OXA-19}</i>	ColKP3, IncFIA (H11), IncFIB, IncFIB, IncFII, IncFII (K), IncR	<i>phoP</i> (4.5), <i>phoQ</i> (5.1)	<i>acrA</i> (3.02), <i>acrB</i> (3.9), <i>tolC</i> (3.6)	<i>ramA</i> (5.3)	<i>kpnE</i> (3.4), <i>kpnF</i> (2)	<i>kpnG</i> (3.4), <i>kpnH</i> (2)	WGS ND	
EN5351/Kp5	ST716/CC716	CTX, CIP-1, TGC-1, SXT	32 R	<i>rmtB</i> , <i>qnrS1</i> , <i>oqxAB</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{TEM-1B}</i> , <i>bla_{SHV-27}</i>	Col4401, IncFIB(K)	<i>phoP</i> (5.7), <i>phoQ</i> (9), <i>pmrA</i> (4.7), <i>pmrB</i> (8.4)	<i>acrA</i> (4.3), <i>acrB</i> (3.9)	<i>ramA</i> (3)	<i>kpnE</i> (3.2), <i>kpnF</i> (3)	<i>kpnG</i> (2.5), <i>kpnH</i> (2.5)	JACMTQ000000000	
EN5359/Kp6	ST37/CC37	CTX, CIP, SXT	32 R	<i>oqxAB</i> , <i>qnrB1</i> , <i>aac(6')-Ib-cr</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{TEM-1B}</i> , <i>bla_{SHV-81}</i> , <i>bla_{OXA-1}</i>	Col4401, FIA (pBK30683), IncFIB (K), IncFII (K)	<i>phoP</i> (4.9), <i>phoQ</i> (7.2), <i>pmrA</i> (5.7), <i>pmrB</i> (10.4)	<i>acrA</i> (4.8), <i>acrB</i> (4), <i>tolC</i> (5.5)	<i>ramA</i> (8.8), <i>soxS</i> (2.2)	<i>kpnE</i> (2), <i>kpnF</i> (2)	<i>kpnG</i> (2), <i>kpnH</i> (2)	JACZCL000000000	
EN5363/Kp7	ST147/CC147	AMK, CTX, FOX, CIP, MEM, SXT	32 R	<i>aac(6')-Ib</i> , <i>oqxAB</i> , <i>qnrS1</i> , <i>aac(6')-Ib-cr</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{NDM-1}</i> , <i>bla_{SHV-11}</i> , <i>bla_{OXA-9}</i>	Col (B5512), Col440I, IncFIA (H11), IncFIB (pQII), IncFII (K), IncR	<i>phoP</i> (5.5), <i>phoQ</i> (6.5), <i>pmrA</i> (6.6), <i>pmrB</i> (10.7)	<i>acrA</i> (2.3), <i>acrB</i> (7.8), <i>tolC</i> (2.1)	<i>ramA</i> (3.4)	<i>kpnF</i> (2)	<i>kpnG</i> (2)	JACSDJ000000000	
EN5387/Kp8	ST101/CC101	AMK, GEN, CZP, FEP, CIP, MEM, SXT	32 R	<i>rmtB</i> , <i>oqxAB</i> , <i>aac(6')-Ib-cr</i> , <i>bla_{NDM-5}</i> , <i>bla_{OXA-48}</i> , <i>bla_{TEM-1B}</i> , <i>bla_{SHV-1}</i>	Col440II, IncFIB (K), IncFII, IncFII (K), IncL, IncR	<i>phoP</i> (4.7), <i>phoQ</i> (7.8), <i>pmrA</i> (10.1), <i>pmrB</i> (7.3)	<i>acrA</i> (4.8), <i>acrB</i> (10.7), <i>tolC</i> (4)	<i>ramA</i> (8.9), <i>soxS</i> (9.7)	<i>kpnE</i> (2), <i>kpnF</i> (3.9)	-	JACSEV000000000	
EN5395/Kp9	ST101/CC101	AMK, GEN, CZP, FEP, CIP, MEM, SXT	32 R	<i>rmtB</i> , <i>oqxAB</i> , <i>qnrS1</i> , <i>aac(6')-Ib-cr</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{NDM-5}</i> , <i>bla_{OXA-48}</i> , <i>bla_{TEM-1B}</i> , <i>bla_{SHV-1}</i>	Col4401, Col440II, IncFIA(H11), IncFIB(K), IncFII, IncFII(K), IncL, IncR	<i>phoP</i> (4.6), <i>phoQ</i> (5.7), <i>pmrA</i> (13.3), <i>pmrB</i> (6.5)	<i>acrA</i> (5.7), <i>acrB</i> (10.8), <i>tolC</i> (4.9)	<i>ramA</i> (9.3), <i>soxS</i> (8.8)	<i>kpnE</i> (2), <i>kpnF</i> (3)	<i>kpnG</i> (2.7)	JACZCM000000000	

Antimicrobial abbreviations: amikacin, AMK; gentamicin, GEN; cefotaxime, CTX; ceftazidime, CTZ; ceftiofur, CEF; ciprofloxacin, CIP; meropenem, MEM; trimethoprim/sulfamethoxazole, SXT; tigecycline, TGC; and colistin, CST.

Abbreviations: incompatibility, Inc; whole genome sequencing not done, WGS ND.

Kp2-Kp4 were clonal, hence, only Kp2 underwent WGS. Its other counterparts i.e., Kp3, Kp4 were assessed through PCR and were included for expression analysis.

EN5219-Susceptible clinical *Klebsiella pneumoniae* strain used as reference and 2^{-ΔΔCt} method was followed for mRNA expression analysis.

Efflux pump inhibitor CCCP caused a decrease in colistin MIC (128–256-fold), while other inhibitors (NMP and PAPN) did not show any change in MIC.

^aThe addition of ‘-I’ after the antimicrobial abbreviation indicates the isolate was intermediate (rather than resistant).

^bData assessed by PCRs only.

^cMean fold change is shown in parentheses after the gene found to be upregulated.

Table 2. Mutational analysis of colistin-resistant *Klebsiella pneumoniae*^a

Isolate	Two-component system and its regulator				Efflux pump genes and outer membrane channel				Regulator and repressor of efflux pumps					Lipopolysaccharide-modifying genes				
	<i>phoPQ</i>	<i>pmrCAB</i>	<i>crdAB</i>	<i>mgrB</i>	RND pump <i>acrAB</i>	SMR pump <i>kpnEF</i>	ABC pump <i>macAB</i>	<i>tolC</i>	<i>acrR</i>	<i>ramR</i>	<i>soxS</i>	<i>marR</i>	<i>pagP</i>	<i>ugd</i>	<i>arn</i> operon	<i>res</i> operon	<i>lpx</i>	<i>lpt</i>
<i>Kp2</i>	<i>phoP</i> : L26Q, R114A, R128S	<i>pmrC</i> : I138V, S257L	WT	C286	WT	<i>kpnE</i> : K112Q	<i>macB</i> : S70T, T116S	N75T, I84V, E205G, N253S, S273N, S278T, S284R, S286N, H291N, N292del, N293T, Q295L, Q296A, Q298N, N300S, A302N	E111D	K91 , S167N	WT	WT	F187I	N354D, T373A	<i>arrB</i> : G47D , I126V, amD; S164P, amT; M114L, V117I, R372K	<i>resC</i> : S35N; <i>resC</i> : V87I	<i>lpxA</i> : S42T; <i>lpxB</i> : D251G ; <i>lpxM</i> : S253G	WT
<i>Kp5</i>	<i>phoP</i> : L26Q, R114A, R128S	<i>pmrB</i> : L213M, T246A; <i>pmrC</i> : I138V, D477E	<i>crdB</i> : E4G	WT	<i>acrA</i> : A379V	WT	<i>macB</i> : S70T, T116S	WT	WT	WT	WT	WT	F187I	D134N, N354A, T373A	<i>arrB</i> : D112A; amD: V53I, I300V; amT: M114L, V117I, H156Q, R372K, D441E	<i>resC</i> : S753N, G843A; <i>resD</i> : E479K	<i>lpxA</i> : S42T, A241T; <i>lpxB</i> : A28E ; <i>lpxM</i> : Q98E; <i>lpxM</i> : A66S , S253G	WT
<i>Kp6</i>	<i>phoP</i> : L26Q, R114A, R128S; <i>phoQ</i> : S174R	<i>pmrB</i> : T246A, R256G ; <i>pmrC</i> : C27F, I138V, Q319R	<i>crdB</i> : I27V	WT	<i>acrA</i> : A188T	WT	<i>macA</i> : N35S; <i>macB</i> : S70T, V130L	N75T, I84V, E205G, N253S, S273N, S278T, I280V, S284R, S286N, H291N, N292del, N293T, Q295L, Q296A, Q298N, N300S, A302N	WT	A19V, I94insH	WT	WT	WT	N354D, T373A, A376V	amA: A18S, A268S, D569E, K635T, V656T, P661S; amB: D112A; amD: E25A , R29K, I300V; amT: M114L, D441E	<i>resC</i> : S753N, G843A; <i>resD</i> : E479K	<i>lpxA</i> : S42T; <i>lpxB</i> : A235G, <i>lpxM</i> : S253G	WT
<i>Kp7</i>	<i>phoP</i> : L26Q, R114A, R128S	<i>pmrB</i> : T246A, R256G ; <i>pmrC</i> : C27F, V50L, A135P, I138V	WT	V326	WT	WT	<i>macB</i> : T116S	T482N	WT	WT	WT	WT	WT	D134N, N354A, T373A	amA: S18A, T185A; amB: D112A; amT: H156Q, R372K	<i>resC</i> : S753N, G843A; <i>resD</i> : E479K	<i>lpxA</i> : S42T; <i>lpxM</i> : S253G	<i>lptD</i> : R586H
<i>Kp8</i>	<i>phoP</i> : L26Q, R114A, R128S; <i>phoQ</i> : T244N	<i>pmrA</i> : A217V; <i>pmrB</i> : T246A; <i>pmrC</i> : G25S, I138V, Q319R	WT	WT	<i>acrB</i> : A305T	<i>kpnF</i> : I39F	<i>macB</i> : S70T, T116S, A611V	N75T, I84V, E205G, N253S, S273N, S278T, S284R, S286N, H291N, N292del, N293T, Q295L, Q296A, Q298N, N300S, A302N	WT	A20D	L60R	S3N	F187I	T373A	<i>arrB</i> : D112A, T283A; amD: V53I; amT: M114L, V117I, G164S , M177I , R372K	<i>resD</i> : S293N, E411Q	<i>lpxA</i> : S42T; <i>lpxM</i> : S253G	WT
<i>Kp9</i>	<i>phoP</i> : L26Q, R114A, R128S	<i>pmrA</i> : Genes not found; <i>pmrB</i> : T246A; <i>pmrC</i> : G25S, I138V, Q319R	WT	C165stop	<i>acrB</i> : A305T	<i>kpnF</i> : I39F	<i>macB</i> : S70T, T116S, A611V	N75T, I84V, E205G, N253S, S273N, S278T, I280V, S284R, S286N, H291N, N292del, N293T, Q295L, Q296A, Q298N, N300S, A302N	WT	A20D	L60R	WT	F187I	T373A	<i>arrB</i> : D112A, T283A; amD: V53I; amT: M114L, V117I, G164S , M177I , R372K	<i>resD</i> : S293N, E411Q	<i>lpxA</i> : S42T; <i>lpxM</i> : S253G	WT

Abbreviations: wild-type, WT; insertion, ins; deletion, del. Polymyxin-susceptible *Klebsiella pneumoniae* MGH 78578/ATCC 700721 (Accession no. CP000647.1) was used as reference for mutational study. ^aOnly genes with mutations have been included in this table, and deleterious mutations according to PROVEAN are shown in bold type. In addition, strains that did not undergo WGS have been excluded from mutational analysis.

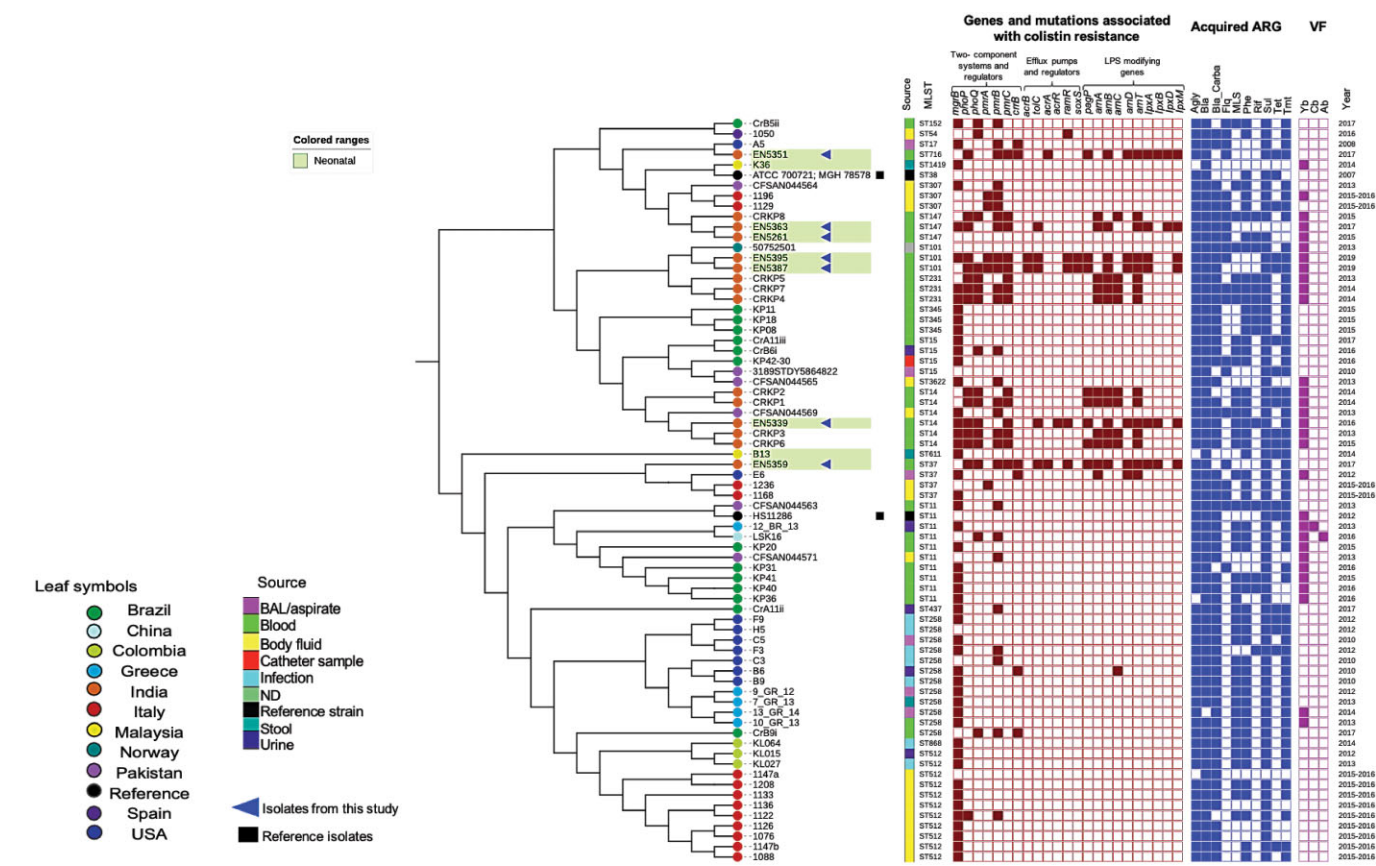


Figure 1. Phylogenomic analysis of colistin-resistant *Klebsiella pneumoniae* strains across the globe reported to harbour chromosomal colistin resistance. The phylogeny consisted of 72 *K. pneumoniae* with 70 colistin-resistant *K. pneumoniae* and two *K. pneumoniae* reference strains. The circles at branch termini are coloured according to the country of origin and the green-coloured ranges indicate isolates from neonates, a blue triangle denotes isolates from this study, and a black square indicates an included reference strain. The source of each isolate is denoted by a coloured box following the dendrogram. Following the MLST text are three gene clusters with presence/absence heatmaps: genes and mutations associated with colistin resistance (dark red); acquired antimicrobial resistance genes (blue); and virulence-associated genes (purple). Abbreviations: multi-locus sequence typing, MLST; sequence type, ST; antimicrobial resistance genes, ARGs; virulence factors, VFs; not determined, ND; aminoglycosides, Agly; β -lactams, Bla; carbapenemases, Bla_Carba; fluoroquinolone, Flq; macrolide-lincosamide-streptogramin B, MLS; phenicol, Phe; rifampicin, Rif; sulphonamide, Sul; tetracycline, Tet; trimethoprim, Tmt; yersiniabactin, Yb; colibactin, Cb; aerobactin, Ab.

study, were analysed in terms of their phylogenetic relatedness (Figure 1). Two aspects were highlighted: (i) association of ST, resistance and virulence genes; and (ii) comparison of genes associated with colistin resistance among the global genomes. Genomes were from different clinical sources and those isolated from blood were from India and Brazil. Study strains were closely related to ColR-Kps from other parts of the world such as the USA, Norway, and other Indian strains. The global collection of ColR-Kps were spread across 21 diverse STs including epidemic clones (ST11, ST258, ST307, ST512) found in Pakistan, China, Greece, Brazil, the USA, and Italy, but the study strains did not belong to these STs. Occurrence of colistin resistance among epidemic clones is particularly worrisome, owing to their propensity to spread. Most of the genomes additionally harboured aminoglycoside- ($n=66$, 94%), β -lactam- ($n=67$, 95%), and carbapenem- ($n=61$, 87%) resistance genes (among others). The majority of the isolates (61/70) analysed, carried at least one carbapenemase gene ($bla_{KPC-2,3}$, bla_{VIM-1} , $bla_{NDM-1,5}$, $bla_{OXA-48,181,232}$), mostly

belonging to epidemic clones. Occurrence of dual carbapenemases such as $bla_{NDM-1,5}$ and $bla_{OXA-48,181,232}$ in ColR-Kp were from India and Pakistan ($n=6$, 3 from this study) while bla_{KPC-3} and bla_{VIM-1} was from Italy. Among the global collection of ColR-Kps, neonatal strains were from this study and Malaysia ($n=2$), but neonatal strains resistant to both carbapenem and colistin were from this unit only. Previous studies of colistin resistance were confined to either TCS genes or *mgrB*, hence, the analysis presented here may be skewed towards these genes. Mutated *mgrB* ($n=51$) was noted across different STs (including study ColR-Kps and epidemic clones), sample types, and countries of origin. Mutations in *phoPQ* and *pmrCAB* were noted in our study, as in other global genomes analysed here.^{1,7-9,11-13} Mutations in study strains were similar to mutations of other Indian genomes.^{8,13} Mutations in LPS-modifying genes or efflux pump genes and repressors were not studied earlier in connection to colistin resistance, but the study strains were found to possess mutations in those genes. Five isolates included in this phylogenomic analysis

had no mutations in colistin resistance genes, including one isolate from this unit (*Kp1*),⁶ indicating the presence of an unknown mechanism of colistin resistance.

Conclusions

Assessment of neonatal septicaemic isolates to understand mechanisms of colistin resistance over a prolonged period is lacking. This study tries to fill in that gap to give a holistic view of the genes and mechanisms involved in colistin resistance. Resistance to colistin is low in the study unit, probably due to limited use of colistin compared with other antimicrobials. Colistin resistance here is due to multiple chromosomal mechanisms that probably work synergistically, and *mcr* was absent. Phylogenomic comparison with global ColR-Kps indicates their clonal diversity, but the colistin resistance mechanisms identified and mutations in associated genes are primarily chromosomally mediated and similar to isolates both at the local and global levels, across several years. This study is the first to report the presence of dual carbapenemases (*bla*_{NDM-5} and *bla*_{OXA-48,181}) in colistin-resistant neonatal *K. pneumoniae*, which is alarming. Improved surveillance and infection control measures will be crucial to contain this resistance.

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Transparency declarations

None to declare.

Supplementary data

Additional Methods information and Figures S1 to S5 are available as [Supplementary data](#) at JAC Online.

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